Math Inquirers

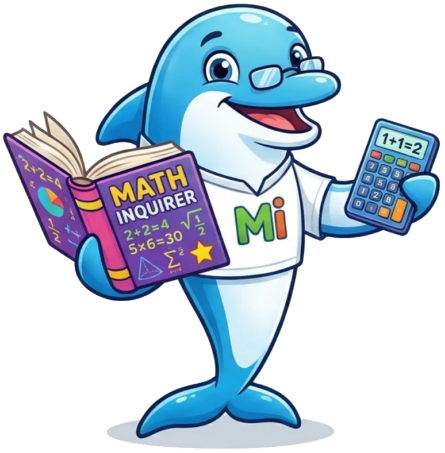
Think. Question. Prove.

Level 1 (ages 6-8) - Term 1

Home Practice Pack

Book 2

Lessons 6 to 10



One sheet for each lesson. Tear out a sheet and do it at home.

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Tens Jump, Ones Hop: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use your bars and squares, or draw the jumps. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

A box has 4 bundles of 10 pencils and 5 single pencils. Ali puts in 3 more bundles of 10. How many pencils are in the box now?

No.2

Start here

Subtract. (a) 86 - 40 = _____ (b) 86 - 4 = _____

No.3

Start here

Ravi and Mei each have 53 stickers. Ravi gives away 3. Mei gives away 30. (a) Ravi has _____ stickers left. (b) Mei has _____ stickers left.

No.4

Think harder

Add 36 + 23 in two ways: (a) jump first, then hop; (b) hop first, then jump. Is the answer the same? Circle Yes or No.

Yes No

No.5

Go further

(a) On spare paper, draw a number line from 0 to 100. Put a coin on 7 and jump it in tens. Write each number. (b) What stays the same?

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 75
4 tens and 3 tens make 7 tens. The 5 loose pencils stay: 75.

2. 46 and 82
40 is 4 tens, a jump back: 46. 4 is a hop back: 82.

3. Ravi 50, Mei 23
Ravi hops back 3: 50. Mei jumps back 3 tens: 23.

4. 59 both ways. Yes, the same.
Jump first: 36 + 20 = 56, then + 3 = 59. Hop first: 36 + 3 = 39, then + 20 = 59.

5. 7, 17, 27 … 97. The ones digit stays the same.
A grown-up checks that each number is 10 more than the one before. The ones digit never changes when you jump in tens.

More practice online (no timers, no scores): ask your tutor for the practice link.

Take Away, Count Up: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use bars and squares, or draw them. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

A jar has 78 sweets. Ali eats 20 of them. How many sweets are left?

No.2

Start here

Subtract: 76 - 43 = _____. Check: _____ + 43 = 76.

No.3

Start here

Subtract: 89 - 36 = _____

No.4

Think harder

Ben has 47 cards. Ali has 71 cards. How many more cards does Ali have than Ben?

No.5

Go further

Pick two page numbers between 20 and 99, such as 38 and 71. How many pages from one to the other? Find it two ways: take away, then count up. Do the two ways match? Circle Yes or No.

Yes No

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 58
78 is 7 tens and 8 ones. Take away 2 tens: 5 tens and 8 ones, which is 58.

2. 33; 33 + 43 = 76
76 take away 40 is 36, and 36 take away 3 is 33. Check: 33 + 43 = 76.

3. 53
89 take away 30 is 59. 59 take away 6 is 53.

4. Ali has 24 more (47 to 50 is 3, 50 to 70 is 20, 70 to 71 is 1)
47 to 50 is 3. 50 to 70 is 20. 70 to 71 is 1. 3 + 20 + 1 = 24.

5. For 38 and 71: 71 - 38 = 33, and counting up gives 2 + 30 + 1 = 33. Both ways match.
A grown-up checks that the child adds the hops and jumps, then adds the answer back to the smaller page number to get the greater one.

More practice online (no timers, no scores): ask your tutor for the practice link.

Balance and Swap: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use two plates and small things, or draw both sides. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Fill in the missing number: 6 + 3 = 5 + ☐

No.2

Start here

Fill in the missing number: 10 + 4 = 9 + ☐

No.3

Start here

Fill in the missing number: 20 + 30 = 40 + ☐

No.4

Think harder

Ravi says 36 + 14 and 34 + 16 have the same total. Do not add. Is he right?

Circle Yes or No. Then write: I know because _____.

Yes No

No.5

Go further

Put 12 small things in a pile of 7 and a pile of 5. Give 1 across and write the new sum, like 6 + 6 = 12. Write all the sums you can find. Many answers are right.

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. **4**
6 + 3 is 9, and 5 + 4 is 9. The 6 became 5 (1 less), so the 3 becomes 4 (1 more).

2. **5**
10 went down by 1 to 9, so 4 goes up by 1 to 5. Both sides make 14.

3. **10**
20 went up by 20 to 40, so 30 goes down by 20 to 10. Both sides make 50.

4. **Yes. 36 went down by 2 to 34 and 14 went up by 2 to 16. Both make 50.**
One number goes down by 2 and the other goes up by 2, so the total stays the same.

5. **Any list that shows the first number going down by 1 and the second going up by 1, for example 7 + 5, 6 + 6, 5 + 7, 4 + 8, 3 + 9, 2 + 10, 1 + 11, 0 + 12.**
Each time one thing moves across, the total stays 12. A grown-up can count the things on both piles to check.

More practice online (no timers, no scores): ask your tutor for the practice link.

Draw and Measure: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use your ruler and spare paper. No ruler? Tell your tutor before the lesson. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Draw a line 8 cm long on spare paper. Start it at the 0 mark of your ruler. Then check: which mark does it end at?

No.2

Start here

A pencil starts at the 0 mark and its tip is at the 17 mark. A crayon is 11 cm long. How many cm longer is the pencil?

No.3

Start here

Measure a spoon, a toy and a crayon in whole cm. Write the 3 lengths in order, shortest first.

No.4

Think harder

A line starts at the 4 mark and ends at the 15 mark. How long is the line? Write: I know because ...

No.5

Go further

Draw a path with 3 straight pieces. The pieces are all different lengths. The whole path is 25 cm long. Write the 3 lengths. A grown-up checks with a ruler.

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. It starts at 0 and ends at 8
A line 8 cm long, drawn from the 0 mark, ends at the 8 mark.

2. 6 cm
The pencil is 17 cm. 17 take away 11 is 6 (count on from 11 to 17: 6 jumps).

3. Many answers are right. Check one by measuring again.
Put one end at the 0 mark and read the mark at the other end. If the end is between two marks, either mark next to it is fine. Then put the numbers in order, smallest first.

4. 11 cm (15 take away 4)
Count the jumps from 4 to 15: 11 jumps. Or take 4 from 15, which is 11. It is not 15 cm, because the line starts at the 4 mark.

5. Many answers are right: any 3 different whole numbers that add up to 25, for example 6 + 8 + 11 = 25.
Choose three different numbers that add to 25, draw each piece with the 0 mark at the start, then measure and add to check.

More practice online (no timers, no scores): ask your tutor for the practice link.

Story Steps: Read It Twice: practice at home

Name _____ Date _____

About 15 minutes, a few days after the lesson. Use real things or draw it. A grown-up may read the questions aloud. Wrong answers are fine: they show what to try next.

No.1

Start here

Ali has 28 marbles. He wins 10 more marbles. Then he gives 5 marbles away. Write one number sentence for each step. Step 1: _____

Step 2: _____ Ali has _____ marbles now.

No.2

Start here

Mei has 14 stickers. Siti has 6 more. How many stickers does Siti have?

No.3

Start here

A box has 40 pencils. 10 pencils are red and 20 pencils are blue. The rest are green. Write one number sentence for each step. Step 1: _____ Step 2: _____ There are _____ green pencils.

No.4

Think harder

Ben has 9 sweets. He has 4 fewer than Kumar. How many sweets altogether?

No.5

Go further

Make up a two-step story with numbers within 100. Write one number sentence for each step. Ask a grown-up to solve it.

Answers for grown-ups

Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 33
Step 1: 28 + 10 = 38. Step 2: 38 - 5 = 33.

2. 20
Siti has more, so add: 14 + 6 = 20.

3. 10
Take away the red and the blue: 40 - 10 = 30, then 30 - 20 = 10.

4. 22
Ben has fewer, so Kumar has more: 9 + 4 = 13. Together: 9 + 13 = 22.

5. Any two-step story with the right sentences. Check each step: the second sentence starts from the answer to the first.
A grown-up checks that each step has one add or one take away, and that the answer sentence matches the question.

More practice online (no timers, no scores): ask your tutor for the practice link.